Kasper Atkinson

kna37@cornell.edu | Engineering Portfolio | +1 (917) 783-4353

Education

Cornell College of Engineering Ithaca, NY
Masters of Engineering, Aerospace Engineering *Anticipated Graduation May 2027*
Bachelors of Science, Mechanical Engineering *GPA: 4.1/4.0*

Work Experience

RTX | Pratt & Whitney East Hartford, CT
Reliability Systems Intern *May 2026 – Present*

- Audited F135's first CMC blade-outer-seal, identifying absent rubbing and thermal-gradient wear modes and authored test-specific severity categorizations to expand DFMEA comprehensiveness from 10 to 14 modes.
- Performed FMECA sensitivity analysis, finding a 25-hour MTBF swing from failure-distribution assumptions.
- Built a Safety Assessment Report module in Cameo, consolidating safety inputs into a reusable MBSE framework.

Imperial College London London, England
Microglider Controls Intern *June 2025 – August 2025*

- Developed MATLAB aerodynamic models for a dragonfly-inspired, tailless tandem-wing 22-gram microglider.
- Simulated mass-shifting pitch control system using PID feedback to achieve sub-10-second stable glide recovery.
- Evaluated 3 trim–linearization methods, weighing speed and accurracy to extract short-period and phugoid modes.

Comarch SA Kraków, Poland
Forecasting Analytics Intern *June 2024 – July 2024*

- Engineered predictive models combining 13 weighted algorithms, improving load forecasts and resource allocation.
- Integrated real-time data pipeline for cloud infrastructure monitoring, reducing reporting lag from 1 hour to seconds.

Leadership and Extracurriculars

Cornell Design Build Fly Ithaca, NY
Chief Engineer & Team Lead *September 2023 – Present*

- Directed all engineering and administrative decisions in four subteams for aircraft development with $35k budget.
- Led team to a 26th-place finish out of 175 international teams, the program's best result in 10 years.
- Ran 40-person team through 9-month competition cycle: exploration, detailed design, manufacturing, testing.
- Led three critical design reviews with a 6-person subteam, producing a technical proposal that ranked 8th.

C-Salt Ithaca, NY
Mechanical Senior Advisor *Aug 2024 – Present*

- Co-founded wave-energy student team, achieving 1st in technical design at DOE's MEC Contest.
- Prepared comprehensive CADs of high-pressure water piston compressor for ocean-water desalination in Guam.

Projects

Aircraft Sizing and Optimization

- Built a convergence trim solver and 5-mode stability analysis to compute mission lap time as the fitness function.
- Modeled propeller via custom BEMT airfoil solver, predicting thrust and torque in off-design and stall conditions.
- Applied genetic algorithm to evolve thousands of high-fidelity aircraft designs, yielding 22% higher mission score.

Morphing Camber Wing Senior Design

- Designed fully enclosed morphing trailing-edge mechanism across 13 iterations, eliminating hinge-gap drag.
- Built a Python-XFOIL pipeline and torque model to size servo actuation across a full 30-degree deflection range.

Nosecone Design and Validation

- Ran ANSYS Fluent CFD to size cooling vents for a PLA motor nosecone, predicting 8 m/s rotating internal flow.
- Predicted an 80F motor temperature steady state, validating to 90F with a 0.7 drag coefficient in wind tunnel.

Skills & Coursework

Skills: Python, MATLAB, Simulink, Excel, SolidWorks, DOORS, DFMA, ANSYS Structural FEA, FMEA, CAM.
Coursework: Mechanical Design, Fluid Mechanics, Materials, System Dynamics, Thermodynamics, Flight Dynamics.
Interests: Aviation history, practical philosophy, learning languages (Polish, French, German), skiing, and vibe coding.